Deployment Method

DEVELOPMENT GUIDE - CODING STANDARDS AND BEST PRACTICES

<Project Name>

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1 IPA FLOWS

1.1 IPA Naming Convention

Below are the recommended naming conventions for ***<ProjectName>***'s IPAs:

- **Outbound/Inbound Interface**
  - ***<ProjectPrefix>***_<INT>_<Source>_<Destination>_<ShortDescription>.lpd
    - e.g. INFR_INT_HCM_SAP_Employee.lpd
- **Workflows/Approval Interface**
  - ***<ProjectPrefix>***_<WF>_<ShortDescription>.lpd
    - e.g. INFR_WF_ApproveDraftRequisition.lpd

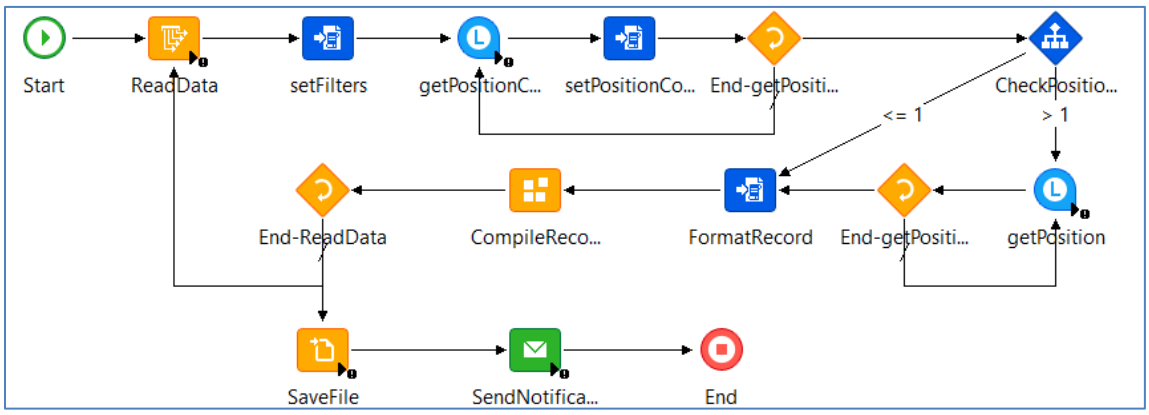
1.2 IPA Rules

The following rules should be followed when creating an IPA for maintability of the flow:

- Never update a standard process flow. If there's a need to update a standard process flow, copy the IPA and rename it (Kindly refer to Section 1.1 IPA Naming Convention). Once the IPA is renamed, update the Service Definition to use the custom flow.
- Declare global variables on Start Node
- Use **CamelCase** or **camelCase** when naming IPA variables and nodes
  - e.g. GetEmployee, EmployeeNumber
- There should be no hardcoded values in the IPA. Constant values must be declared in the System Configuration.

String	vFileDirectoryFTP	_configuration.BusinessSolver.FTPFileDirectory

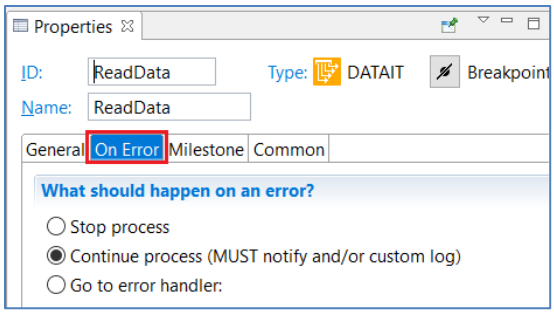
- Use descriptive names for variables and functions:
- Use descriptive names for Nodes:
  - Make sure the node names are equivalent to node IDs, this will make the debugging of IPAs much easier.



- Code Reusabilty - IPA Functions should be declared as early as possible to make it reusable. Avoid setting the function inside the loop.
- IPA flows should have error handling. Kindly refer to Section 1.3 IPA Error Handling.

1.3 IPA Error Handling

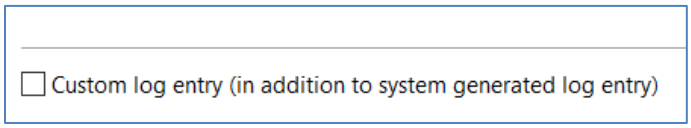
IPA Error Handling can be accessed through the “On Error” tab of the IPA Node:



Each error handling option can be associated with an email notification or a work unit log message:

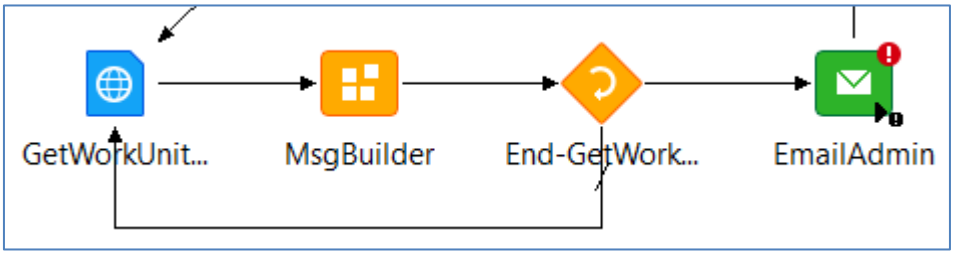
- Notify
  - This will send an email notification with the custom error message

- Custom Log Entry
  - This will log a custom error message in the work unit log

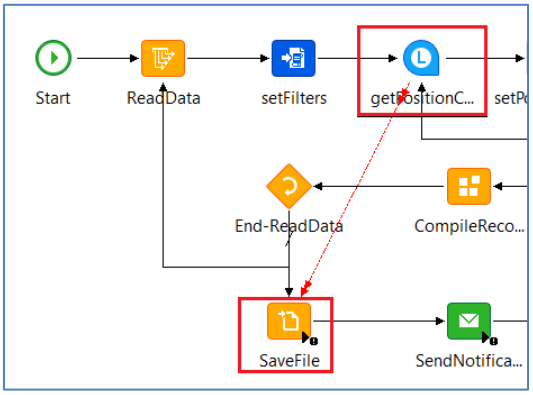


Below are the different error handling options, and when to use them:

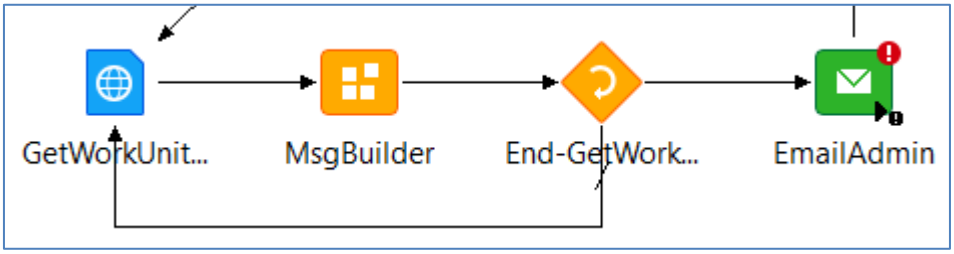
- Stop Process
  - Use this option if the IPA should be terminated immediately when there’s an error in the IPA node.
  - When using this node, make sure that the “Notify” option is selected to notify the necessary recipeints of the error.
- Continue Process
  - Use this option if the IPA should still continue on the usual process flow regardless of the error encountered.
  - When using this node, make sure that the “Custom log entry” option is selected.
  - Reuse GetWorkUnitErrors (can be found on standard workflow IPAs) to compile the errors from the WorkUnit and send it to HR Admin/IT.



- Go to Error handler
  - Use this option if the IPA should still continue the process but proceeds on a different node regardless of the error encountered.



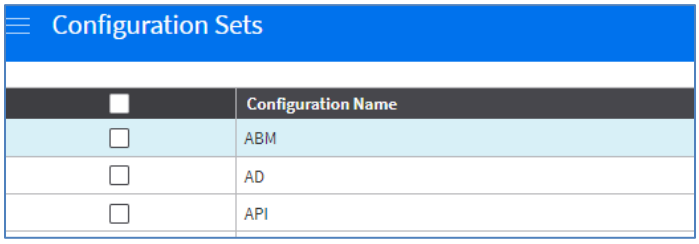
  - When using this node, make sure that the “Custom log entry” option is selected.
  - Reuse GetWorkUnitErrors (can be found on standard workflow IPAs) to compile the errors from the WorkUnit and send it to HR Admin/IT.



1.4 System Configuration Rules

The following rules should be followed when creating a System Configuration:

- Each vendor should have its own system configuration sets on Process Server Administrator -> Configuration -> System Configuration:



- Variables in each System Configuration Set should be in CamelCase
  - e.g. FileDirectory, FileName
- Each variable should include a description containing the name of the IPA where it is being used. This will make it easier to identify which fields should be updated for a specific IPA.

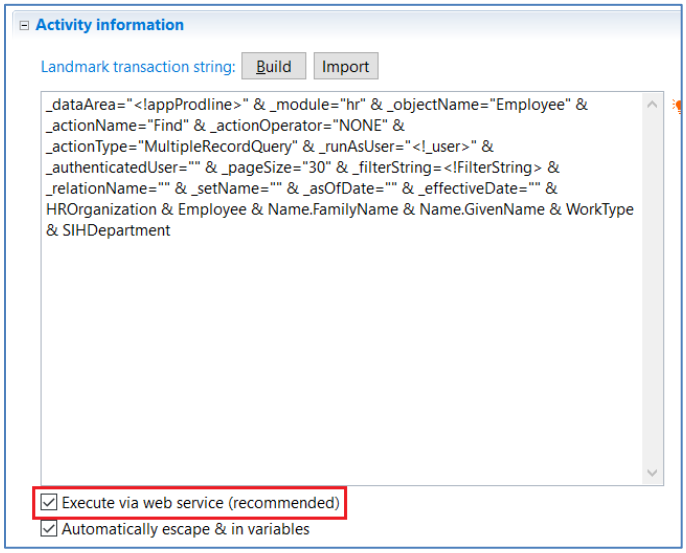
1.5 Performance Consideration

The following should be considered when creating an IPA to avoid any performance issues:

- Use webservice on landmark transactions to avoid memory alert error.

  Note : A generic landmark user with all the required security roles must be created and used for the _runAsUser.

  <ProjectName>'s _runAsUser : ***<Insert Project's generic Actor here>***



- Always use unique key fields (ex. Employee, Position, WorkAssignment.. etc.) when filtering records. Filters should never be based from names of fields that may contain common values.
  - Use _filterString when the actionType is a MultipleRecordQuery
  - If the actionType is a SingleRecordQuery, make sure that all key fields are populated.

2 FILE CREATION UTILITY

2.1 Naming Convention

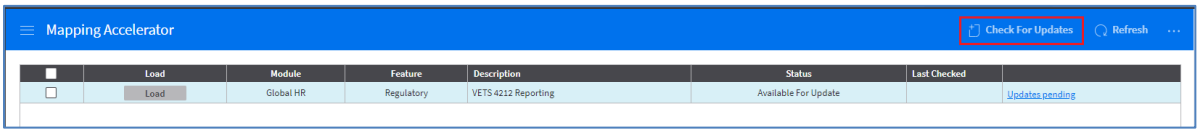
Below are the recommended naming convention for ***<ProjectName>***'s FCU Setup:

- ***<ProjectPrefix>***_<FCU>_<Source>_<Destination>_<ShortDescription>
  - INFR_FCU_HCM_ADP_EmployeePosition

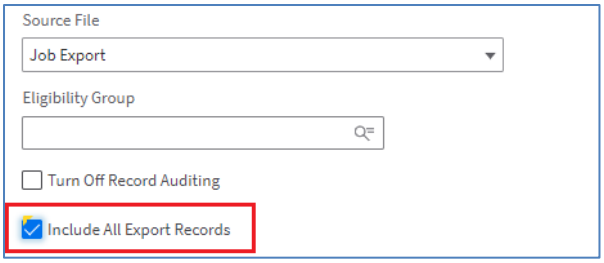
2.2 FCU Rules

The following rules should be followed when creating an FCU Setup:

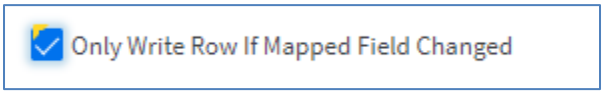
- Instead of using Data Mapping for Data manipulation, use derived fields to make it reusable to other FCU setups. Kindly refer to section 4. Configuration Console for guidelines on creating a configuration console item.
- Instead of Java Format and Data Mapping, use derived fields to format the field value. Derived fields makes the processing of records faster than when using Java Format and Data Mapping.
- Use ION instead of IPA for transferring files. Only use IPA if data manipuation is needed.
- Reuse FCU setups available on Delivered Content (Administrator -> Utilities -> Interfaces -> File Creation Utility -> Delivered Content).
  - Click "Check for Updates" to load available FCU setups



- For records coming from export business class, use "Include All Export Records" option if you want to include records in the export file regardless of their changed fields.



- Use "Only Write Row If Mapped Field Changed" option inside the row mapping if the record should only be generated when a mapped field inside the row was changed.



3 ION

3.1 Naming Convention

Below are the recommended naming convention for ***<ProjectName>***'s ION Items:

- **BOD**
  - ***<ProjectPrefix>***_<BOD>_ <ShortDescription/BusinessClass>.xml
    - INFR_BOD_Personnel.xml
- **Data Flow**
  - ***<ProjectPrefix>***_<DataFlow>_<Source>_<Destination>_<ShortDescription>
    - e.g. INFR_DataFlow_HCM_SAP_Employee
- **Connection Points**
  - ***<ProjectPrefix>***_<CP>_<Vendor>
    - INFR_CP_ADP
- **File Mapping**
  - ***<ProjectPrefix>***_<FM>_<Source>_<Destination>_<ShortDescription>
    - e.g. INFR_FM_HCM_ADP_Employee.lpd

4 CONFIGURATION CONSOLE

4.1 Naming Convention

Below are the recommended naming conventions for ***<ProjectName>***'s Configuration Console items:

- Use CamelCase when naming Configuration Console items.
- Use descriptive names.
- Always include the client's acronym as a prefix/suffix when naming custom fields, forms, lists, actions, relations, and etc.

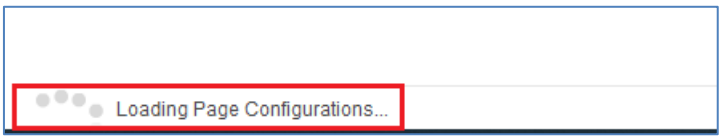
Configuration Console items Format:

- ***<ProjectPrefix>***<VariableName>
    - INFTestVariable

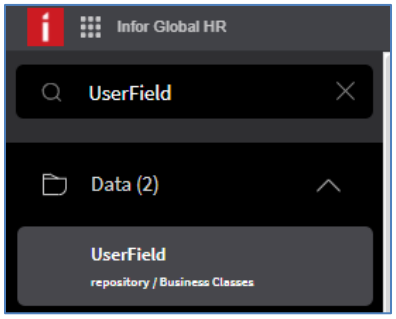
4.2 Configuration Console Rules

The following rules should be followed when creating Configuration Console items:

- Add the create configuration console item in the project's spreadsheet. This will avoid duplicate items.
    - ***<ProjectName>***'s Spreadsheet directory: ***<Insert Project's spreadsheet location>***
- Always check the Configuration console item spreadsheet before creating a new configuration console item.
- When using the Rich Client, make sure that prior doing configurations, everything has been loaded to avoid discrepancies. This is found on the lower left portion of RichClient.



- Safest way to update custom items is on the Business Class (UserField, UserBusinessClass, UserAction.. etc)



5 SECURITY

5.1 Naming Convention

Below are the recommended naming conventions for ***<ProjectName>***'s Security items:

- Use CamelCase when naming Security items.
- Use descriptive names.
- Always include the client's acronym as a prefix/suffix when naming Security Roles and Securit Classes.

Configuration Console items Format:

- Security Class : <VariableName>_***<ProjectPrefix>***
  - ABMProcessingView_INF
  - GHRGenEmpBankDetails_INF
- Security Role : <VariableName>_***<ProjectPrefix>***
  - Employee_INF
  - HRGeneralist_INF

5.2 Security Rules

The following rules should be followed when creating Security items:

- Do not edit the Standard (_ST) security classes or roles.
  - Always make a copy of the security class and replace _ST with the project's initials (ex. DirectSupervisor_INF).
- Every field - custom or whatnot can be used in your security rules.
  - Use this to your advantage to shorten conditions or if the config console won't allow 2 conditions in a form.
  - Use derived fields often and use it in your security rules when possible
  - Use derived fields on actions on composite forms that won't go away even if you restricted it in security. (Some clients won't like to see errors popping up if they clicked a button)
- On editing Security classes in a role – always add your custom security class before you delete the standard security class - for custom roles only.
- Avoid sharing the same security class on a custom role – this might get confusing if the requirements change along the way. It's more work but future you will thank you for doing this.
- Create an excel file mapping out your RoleSecurityClass design and update it often -what you have changed, what you have added, what you have replaced.
  - ***<ProjectName>***'s Spreadsheet directory: ***<Insert Project's spreadsheet location>***
- All webapps, menu, menuitems security rules are in most likely in *webapp* security classes. Keep it that way.
- If you can, avoid adding new security classes to ST roles. Exceptions are the following: User Business Class, User Action - or any Config Console item that will ask you what roles it should be applicable.
- For workflow approval issues related to security on Landmark8000, consult with a security consultant.

- Careful with adding customizations with HRGeneralistActorOrgUnit roles - this role has a very sensitive integrity - better consult with your security consultant if this role has customizations.
- When doing Custom Security Roles, focus on security configurations only. As much as possible, avoid updating LPLs. We can restrict fields/actions on Security.